EE3-09
EE9-CS1-1

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2017

EEE/EIE PART III/IV: MEng, BEng and ACGI

Corrected copy

CONTROL ENGINEERING

Wednesday, 12 December 9:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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CONTROL ENGINEERING

1. Consider a thermal system composed of three bodies of thermal capacitance $\kappa_1 > 0$, $\kappa_2 > 0$, and $\kappa_3 > 0$ and temperature T_1, T_2, and T_3, respectively. Body 1 is in contact with a body at constant temperature T_0 and with Body 2; Body 2 is in contact with Bodies 1 and 3; Body 3 is in contact with Body 2 and receives a heat flow u from an external source. The heat exchange coefficients between the different bodies are $k_{01} \geq 0$, $k_{12} \geq 0$, and $k_{23} \geq 0$, respectively. The system can be described by the equations

$$\begin{aligned} \kappa_1 \dot{T}_1 &= -k_{01}(T_1 - T_0) + k_{12}(T_2 - T_1), \\ \kappa_2 \dot{T}_2 &= -k_{12}(T_2 - T_1) + k_{23}(T_3 - T_2), \\ \kappa_3 \dot{T}_3 &= -k_{23}(T_3 - T_2) + u. \end{aligned}$$

a) Defining the state variables $x_1 = T_1 - T_0$, $x_2 = T_2 - T_0$, and $x_3 = T_3 - T_0$; the input signal u; the output $y = x_1$ and recalling that T_0 is constant, write a state space description of the system, that is the matrices A, B and C of the system. [6 marks]

b) Let $\kappa_1 = \kappa_2 = \kappa_3 = 1$ (J/K) and $k_{01} = k_{12} = 1$ (W/K).

i) Study the stability of the system as a function of $k_{23} \geq 0$. (Hint: Use Routh test.) [4 marks]

ii) Compute the observability matrix of the system and determine for which values of $k_{23} \geq 0$ the system is observable. [4 marks]

c) Let $k_{23} = 0$. Show, using only the equations describing the system, that the system is neither controllable nor observable. Is the system stabilizable and/or detectable? [6 marks]

2. The zeros of a linear, time-invariant, single-input single-output, system described by the equations

$$\sigma x = Ax + Bu, \qquad y = Cx + Du,$$

with state $x(t) \in \mathbb{R}^n$, input $u(t) \in \mathbb{R}$, and output $y(t) \in \mathbb{R}$, are the complex numbers ζ such that the determinant of the *system matrix*

$$Z(\zeta) = \begin{bmatrix} \zeta I - A & B \\ C & -D \end{bmatrix}$$

is zero. In what follows, we refer to the system above as the "open-loop" system.

a) Assume, for simplicity, that $D = 0$. To study the effect of state feedback on zeros let

$$u = Kx + v.$$

Write the equations of the resulting closed-loop system and show that this has exactly the same zeros as the open-loop system.
(Hint: Write the *system matrix* of the closed-loop system as the product of the *system matrix* of the open-loop system and a constant matrix.)

[8 marks]

b) Assume, for simplicity, that $D = 0$. To study the effect of output injection on zeros consider the modified system (that is a system in which the output has been injected in the first equation):

$$\sigma x = Ax + Bu + Ly, \qquad y = Cx,$$

and show that this has exactly the same zeros as the open-loop system.

[6 marks]

c) Assume, for simplicity, that $D = 0$. Consider, finally, the effect of feedforward, that is replace the output y with a new output $\tilde{y}$ obtained by feeding the input into the output, yielding the system

$$\sigma x = Ax + Bu, \qquad \tilde{y} = Cx + \delta u,$$

with δ constant and different from zero.
To study this case consider the open-loop system

$$\sigma x = x + u, \qquad y = x,$$

with $x(t) \in \mathbb{R}$, $u(t) \in \mathbb{R}$, $y(t) \in \mathbb{R}$, and the system obtained using feedforward, that is

$$\sigma x = x + u, \qquad \tilde{y} = x + \delta u,$$

with $\tilde{y}(t) \in \mathbb{R}$. Show that the open-loop system does not have any zero and that the system with feedforward has a zero for all $\delta \neq 0$. What happens to the zero when $\delta \to 0$? State your conclusions concerning the effect of feedforward on zeros.

[6 marks]

3. Consider the linear, discrete-time, system

$$x_1^+ = x_1 + u, \qquad x_2^+ = 2x_1 + ax_2 + u,$$

with $x_1(t) \in \mathbb{R}$, $x_2(t) \in \mathbb{R}$, $u(t) \in \mathbb{R}$ and a a constant parameter.

a) Study the reachability, controllability and stabilizability properties of the system as a function of the parameter a. [4 marks]

b) Let $a = 0$. Explain why it is possible to steer the state of the system from $x_1(0) = x_2(0) = 1$ to $x_1(2) = x_2(2) = 2$ and determine an input sequence achieving the transfer. [6 marks]

c) Design, if possible, a state feedback control law $u = Kx$ such that both eigenvalues of the closed-loop system are located at 0. (Note that the state feedback gain K is a function of a.) Determine for which values of a the feedback gain K is well-defined and comment on your result. [6 marks]

d) Let $a = -1$. Determine all states that can be reached from $x_1(0) = x_2(0) = 0$, regardless of the number of steps. [4 marks]

4. Consider the linear, continuous-time, time-invariant, system described by the equations

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = x_3, \qquad \dot{x}_3 = d, \qquad y = x_1 + x_2 + x_3,$$

with state $x = [x_1, x_2, x_3]'$, with $x_i(t) \in \mathbb{R}$, for $i = 1,2,3$, disturbance input $d(t) \in \mathbb{R}$ and output $y(t) \in \mathbb{R}$.

a) Show that the system is observable. [2 marks]

b) Assume that the disturbance input is measurable. Design an observer for the state of the system such that the eigenvalues of the estimation error system are located at -1, -2, and -3. Write, explicitly, the equations of the observer. [6 marks]

c) Suppose now that the disturbance input is not measurable and that one wishes to estimate only the signal $w = x_1$. This objective can be achieved building a reduced order observer in steps.

i) Write the equations of the system for which one wishes to design the observer in the form

$$\dot{x} = Ax + Dd, \qquad y = Cx, \qquad w = Hx.$$

Write explicitly the matrices A, D, C and H. [4 marks]

ii) Define the observer dynamics as

$$\dot{z} = Nz + Ly, \qquad \hat{w} = Fz,$$

with $z(t) \in \mathbb{R}^2$ and $\hat{w}(t) \in \mathbb{R}$. Show that the matrices $V \in \mathbb{R}^{2\times 3}$ and $L \in \mathbb{R}^{2\times 1}$ defined as

$$V = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \qquad L = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

are such that

$$VD = 0 \qquad NV - VA + LC = 0,$$

for some N. Determine N explicitly. [4 marks]

iii) Show that the error signal

$$e = z - Vx$$

is such that $\dot{e} = Ne$ and that e exponentially converges to zero, that is the system $\dot{e} = Ne$ is asymptotically stable. [2 marks]

iv) Using the results of part c.iii) determine F such that $\hat{w}$ provide an asymptotic estimate of x_1 regardless of d. [2 marks]